Amazon – Global E-Commerce Intelligence Suite – Inventory & Supply Chain Management

Report prepared and analytics conducted by Soham S. Amburle | 26 – 28 February 2026

The Inventory & Supply Chain Management Dashboard provides an executive and operational view of inventory flow, fulfilment efficiency, delivery reliability, and cost management within a global e-commerce business model inspired by Amazon. Built using a synthetic dataset of 100,000 transaction-level records, this dashboard focuses on monitoring inventory movement trends, product category distribution, delivery performance across regions, fulfilment cost allocation, and order status distribution.

Designed in Power BI using a structured star schema data model, the dashboard enables executive leadership, supply chain teams, and operations managers to evaluate operational efficiency, identify bottlenecks, and optimize inventory and logistics strategy through dynamic filtering and interactive visual storytelling.

Through KPI cards, slicers, trend charts, bar/column charts, pie charts, and tree maps, the dashboard delivers a consolidated and executive-ready perspective of supply chain operations.

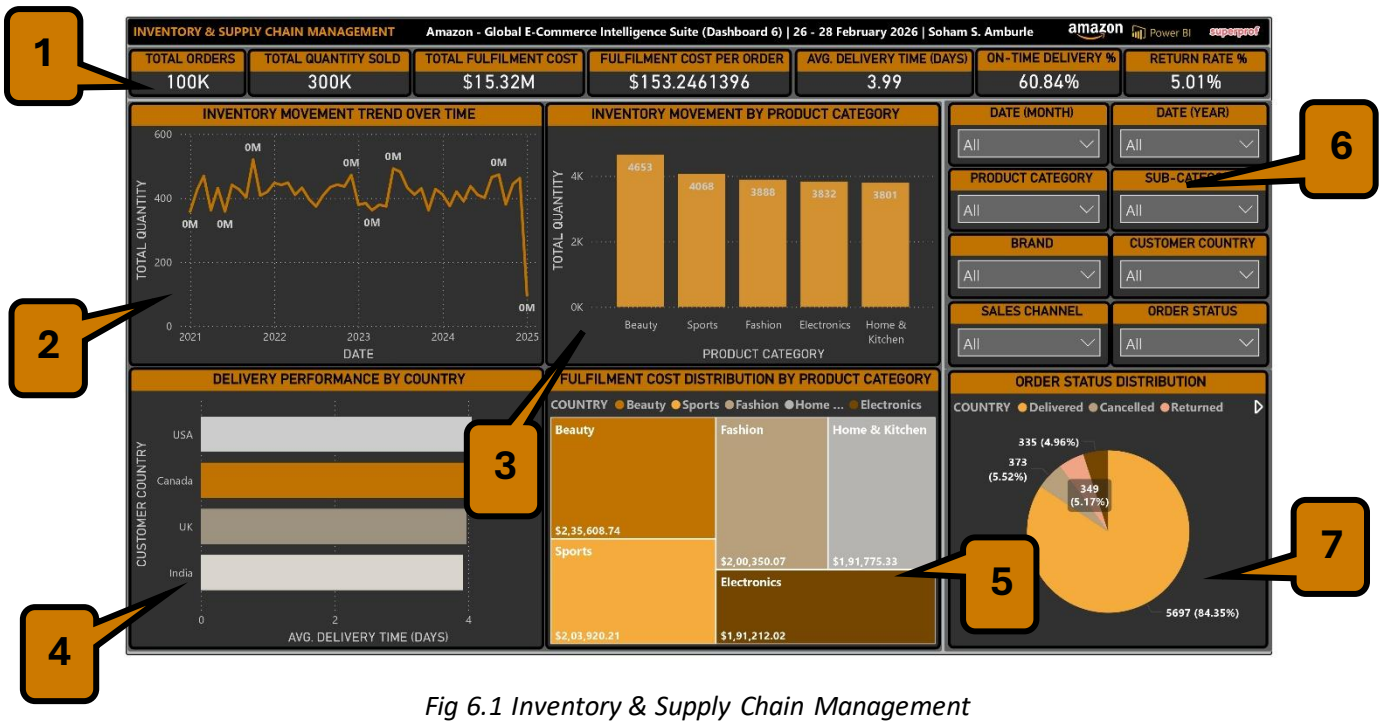

Fig 6.1 Inventory & Supply Chain Management

Executive Storyline

The dashboard follows a structured operational analytics narrative:

- Assess overall order volume and inventory movement
- Measure fulfilment cost efficiency and cost per order
- Evaluate delivery performance and average delivery time
- Compare on-time delivery rates across regions and product categories
- Analyze return rates to monitor reverse logistics
- Identify high-cost product categories and optimize resource allocation

Data Disclaimer: This project uses a fully synthetic dataset of 100,000 records for demonstration and testing only; no real-world data was used.

This layered structure ensures stakeholders can move from high-level KPIs to detailed operational insights.

Visuals and Description

1. KPI Cards (Executive Marketing Snapshot)

Seven KPI cards provide an immediate summary of operational performance:

1. Total Orders: Overall orders processed within the selected filter context
2. Total Quantity Sold: Units sold across all product categories
3. Total Fulfilment Cost: Total cost incurred for order fulfilment
4. Fulfilment Cost per Order: Average cost to fulfil a single order
5. Average Delivery Time (Days): Average days between shipment and delivery
6. On-Time Delivery %: Share of orders delivered within expected timeframe
7. Return Rate %: Percentage of orders returned

These KPIs establish order scale, inventory movement, cost efficiency, and delivery reliability at an executive level.

2. Inventory Movement Trend Over Time (Line Chart)

X-Axis: Order_Date (Month)

Y-Axis: Total Quantity Sold

Unit: Units

Purpose: Shows inventory flow trends and operational demand patterns over time.

3. Inventory Movement by Product Category (Clustered Column Chart)

X-Axis: Product_Category

Y-Axis: Total Quantity Sold

Legend (optional): Sub-Category

Unit: Units

Purpose: Highlights movement volume by product category to identify high-turnover items.

4. Delivery Performance by Country (Clustered Bar Chart)

X-Axis: Avg Delivery Time (Days)

Y-Axis: Customer_Country

Color: On-Time Delivery %

Unit: Days / %

Purpose: Compare delivery efficiency across countries and pinpoint regions needing improvement.

5. Fulfilment Cost Distribution by Product Category (Tree Map)

Group: Product_Category

Values: Total Fulfilment Cost

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Unit: Currency ($, display units in Millions)

Purpose: Visualize cost concentration across product categories for operational cost optimization.

6. Interactive Filtering Controls (Slicers)

The dashboard includes the following dynamic slicers:

1. DATE (Month)
2. DATE (Year)
3. Product Category
4. Sub-Category
5. Brand
6. Customer Country
7. Sales Channel (Web / Mobile App)
8. Order Status

These enable multi-dimensional operational and inventory analysis.

7. Order Status Distribution (Pie Chart)

Values: Count of Orders

Legend: Order_Status

Unit: Count of Orders

Purpose: Provides visibility into delivered, returned, and cancelled orders for reverse logistics monitoring.

Business Value

This dashboard enables stakeholders to:

- Monitor inventory and supply chain KPIs in a consolidated executive view
- Compare product categories and regional performance
- Identify high-cost categories and delivery bottlenecks
- Track returns and operational efficiency over time
- Make data-driven supply chain, inventory, and fulfilment decisions

By integrating KPI cards, slicers, trend analysis, geographic and category-level visuals, the dashboard delivers a structured and enterprise-ready supply chain intelligence solution.

Tools & Data

- Visualization Tool: Power BI
- Dataset: Fully synthetic global e-commerce dataset
- Record Count: 100,000 transaction-level records
- Time Period: 2021–2024
- Model Design: Star Schema (Fact_Orders connected to Date, Customer, Product, and Channel dimensions)
- Purpose: Inventory & supply chain performance analytics and enterprise BI portfolio demonstration

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